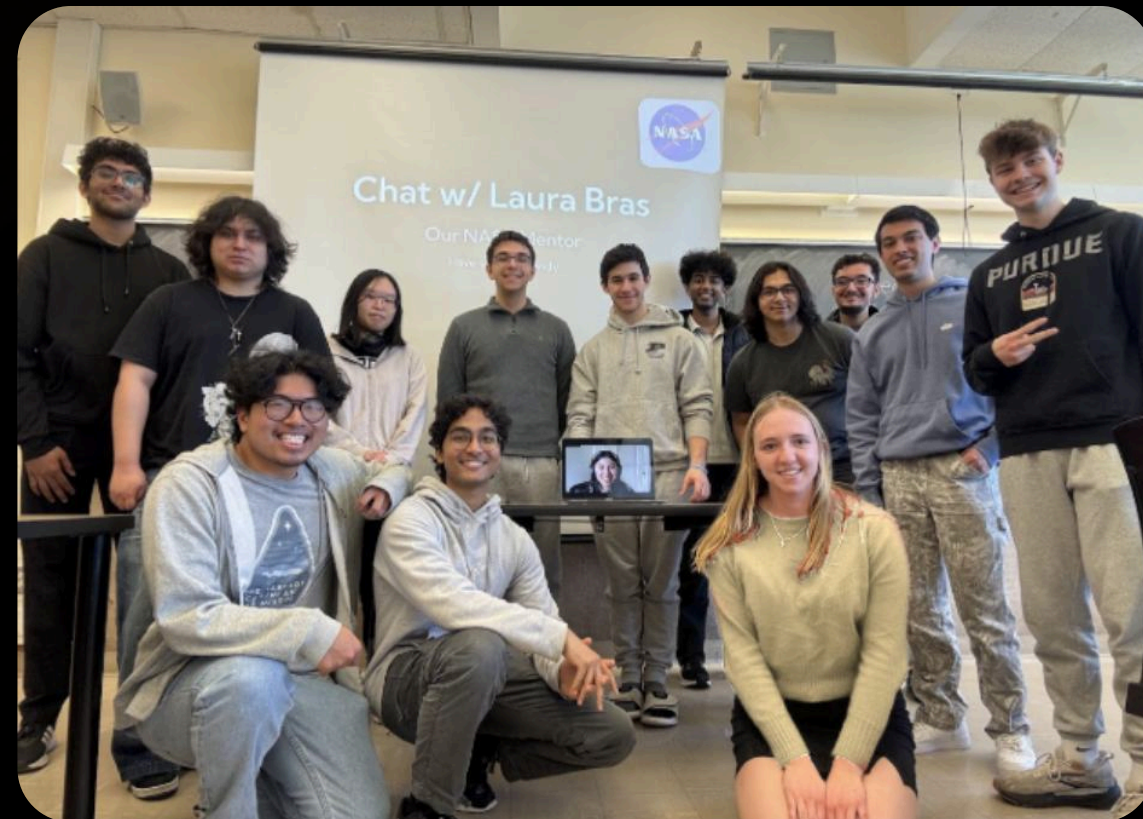


VISOR - 2026



Visual Interface for Space Operations & Reconnaissance



OBJECTIVES

To **GUIDE** our design process, we follow the **Rules of Human Spaceflight** which are listed below in order of importance:

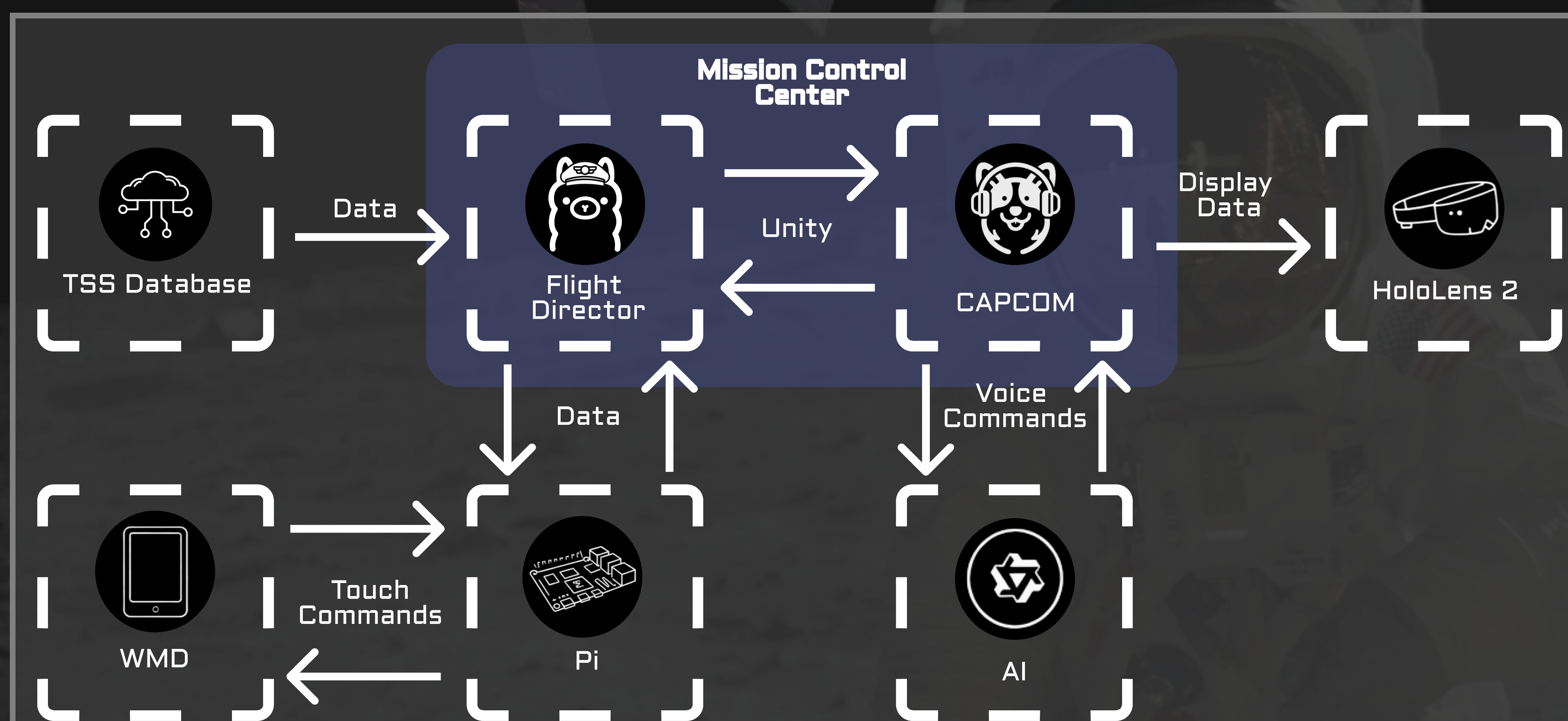
1. **Safety** of the **CREW**
2. **Safety** of the **EQUIPMENT**
3. **Success** of the **MISSION**

VISOR COMPANION

VISOR is a **voice-interactive AI assistant** designed to **support** astronauts in **REAL-TIME**

- Wake phase: "Hey **VISOR**"
- Uses **Retrieval-Augmented Generation** (RAG) Framework for **contextual & accurate** data
- Enable **hands-free interactive** and **reduce cognitive load** in **EVA**s

ASTRONAUT INTERFACE



ASTRONAUT INTERFACE DIAGRAM

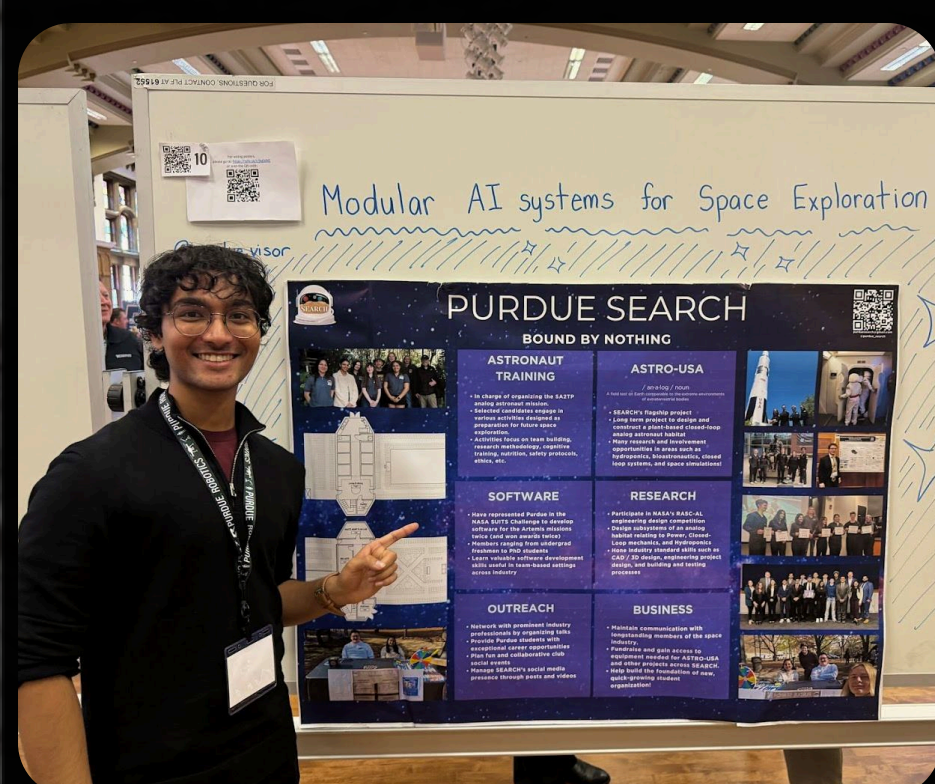


HMD & WMD INTERFACE

The **TSS Database** flows into the **Flight Director**, our **CENTRAL HUB**. This information is then sent to the **Raspberry Pi**, which connects to the **Wrist-Mounted Display (WMD)** for real-time **TOUCH INTERACTION**. At the same time, the Flight Director **COMMUNICATES** with **CAPCOM**, who manages the **Head-Mounted Display (HMD)** and **interfaces** with **VISOR**. **VISOR PROCESSES** these **inputs** and **DELIVERS** responses to the **HMD**, giving the astronaut **REAL-TIME** information.

OUTREACH EVENTS

400+ in-person connections
7800+ virtual connections



Purdue Inaugural Robotics Day

Presented the applications of our AI system at an annual robotics symposium



Attica Middle and High School Visit

Advocated the possibilities in space within the STEM field to 6th - 9th graders



In-Class Guest Lecture

Introduced engineers to software and materials science applications within space

Amber Joneleit • Anand Shanker • Anthony Dierkes • Azeem Ehtisham • Benji Emini • Chawin Mingsuwan • Erik Gutierrez • Faisal Alsayyari • Jason White • Madelyn Christiansen • Paul Greenberg • Peter Zakariya • Sathvik Valluri • Sean VandenBussche • Sejal Verma • Swastik Patel • Trinity Turvey • Zach O'Connell